Does the stop decision reduce to the binding constraint?

Pit-loss cost and the tyre/fuel boundary across 205 endurance race-seasons

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The question

The tank sets a floor on the number of stops; tyre degradation can in principle justify one more than that floor. My question is whether that is settled by the price of the binding constraint alone — once a stop costs enough, no car's degradation is steep enough to pay for it — rather than by the machinery. Your competitor-aware work prices charge and the stops it forces; this asks whether the price of the constraint, not the vehicle, sets the regime.

How the two quantities are measured

Pit loss, per green-flag stop, in-lap and following out-lap both under green:

$$\text{loss} = t_{in} + t_{out} - 2 \times \text{that driver's median green pace in that race}$$

so it carries pit-lane transit and stationary time together, reported per class as a median. Fuel range F is the ceiling of the 0.9 quantile of laps between pit visits, not the maximum, which is contaminated by cars that had stopped racing.

Tyre-limited, by exact dynamic program. For race length L, green pace p, fitted net slope β and pit loss π, a plan partitions L into stints $(n_1 \dots n_{m+1})$ with m stops, costing

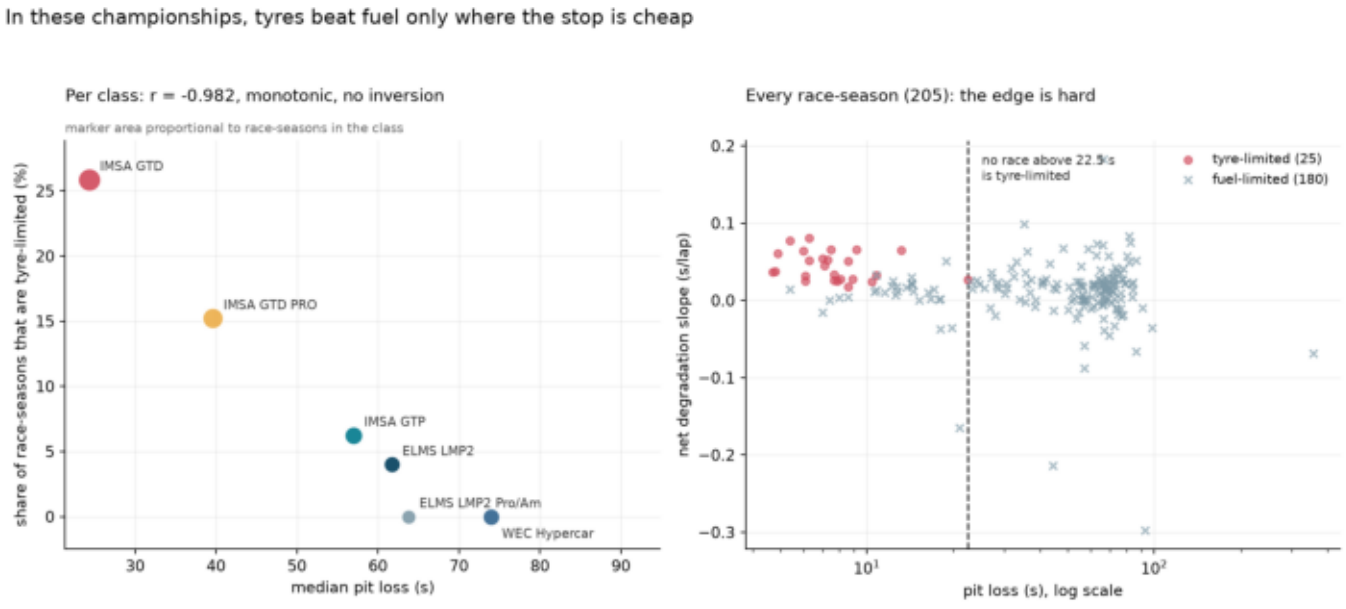
$$T = m\cdot\pi + \Sigma_j \left(n_j\cdot p + \beta\cdot n_j(n_j+1)/2 \right) \qquad \text{subject to } n_j \le F$$

minimised exactly for each m from the fuel minimum $\lceil L/F \rceil - 1$ upward; tyre-limited means the time-optimal m exceeds that minimum. The triangular sum is exact rather than the $\beta\cdot n^2/2$ integral — they differ by $\beta\cdot n/2$, about a second over a forty-lap stint, and that is the only quantity weighed against π. β is fitted per race-season with driver–race fixed effects, as a net slope: the source carries no compound label.

The correlation

class	median pit loss	tyre-limited	race-seasons
IMSA GTD	24.4 s	25.9 %	58
IMSA GTD PRO	39.6 s	15.2 %	46
IMSA GTP	57.0 s	6.2 %	32
ELMS LMP2	61.7 s	4.0 %	25
ELMS LMP2 Pro/Am	63.8 s	0.0 %	17
WEC Hypercar	74.0 s	0.0 %	27

r = -0.982, 95% CI [-0.986, -0.745], monotonic across all six classes with no inversion. The interval is bootstrapped over races within class (205 races, 10,000 replicates), not over the six class summaries — six fixed championships are not a source of variation. At race level, 150 of 205 race-seasons sit above 22.5 s of pit loss and 0 of those is tyre-limited, across every class (25 of 205 are tyre-limited in total).



Where this evidence is thin

The 22.5 s edge is a maximum, and one race puts it there: IMSA GTD at Indianapolis 2025. The next tyre-limited race sits at 13.2 s, so removing that one race moves it by 41%. I report no interval for it: a maximum sits on the boundary of its own support, so the bootstrap is degenerate above it and understates uncertainty below. The rule is not thin — the constant is.

What I would value your view on

Whether a correlation across six class summaries is the right statistic at all. The alternative I see is a per-race model with pit loss as covariate and tyre-limited as outcome, using all 205 race-seasons instead of six points, at the cost of modelling the within-class dependence those summaries absorb.